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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2020 (New Course)

Code: 104504 | ***Subject: Linear Integrated Circuits and Applications*** | ***Time: 03 Hours*** | ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are EIGHT questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Answer any seven of the following short answer-type questions : **[2×7=14]**

(a) Write two requirements to be met for a good current source.

(b) Why VCO is also called as v-to-f converter?

(c) What is a switched capacitor filter?

(d) What output voltage would be produced by a D/A converter whose output range is 0 to 10 V and whose input binary number is 0110 for a 4-bit DAC?

(e) What is a power amplifier?

(f) What is the purpose of connecting a capacitor at the input and the output side of an IC voltage regulator?

(g) Mention any two applications of PLL.

(h) For an n-bit flash-type A/D converter, how many comparators are required?

(i) What is meant by frequency synthesizing?

(j) Why are integrators preferred over differentiators?

Q.2 Solve both questions : **[7+7=14]**

(a) With suitable circuit diagram, explain the operation of voltage reference circuit using temperature compensation.

(b) What are the different methods used in fabrication of integrated resistors? Explain any one method with suitable diagrams.

Q.3 Solve both questions : **[7+7=14]**

(a) Draw and explain the circuit of a voltage to current converter if the load is (i) floating and (ii) grounded.

(b) Define and explain slew rate. What is full-power bandwidth? Also explain the methods to improve the slew rate.

Q.4 Solve both questions : **[7+7=14]**

(a) Explain the working of a anti logarithmic amplifier using suitable op-amp circuit diagram and supporting equations.

(b) Draw the functional diagram of 723 regulators and explain its working.

Q.5 Solve both questions : **[7+7=14]**

(a) Draw the block diagram and explain the principle of working, characteristic and applications of a frequency shift keying (FSK) demodulator.

(b) Explain the operation of frequency to voltage converter with suitable circuit diagram.

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Q.6 Solve both questions : **[7+7=14]**

(a) Explain the working of R-2R ladder D/A converter with circuit diagram.

(b) Explain the oversampling A/D converter with functional block diagram.

Q.7 Solve both questions : **[7+7=14]**

(a) Explain with block diagram how PLL is used as frequency modulation (FM) detector.

(b) Design a VCO having the maximum range of 1 kHz. Assume power supply V_{cc}= 15V. Required pulse width should not exceed 100 μs.

Q.8 Solve both questions : **[7+7=14]**

(a) Design an astable multivibrator circuit which will flash the electric bulb such that its on time will be 4 seconds and off time will be 2 seconds.

(b) Explain the working of Schmitt trigger using 555 timer.

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